

SCOPE OF WORK

Furnish and Install 30 kW Standby Generator System

1. PROJECT DESCRIPTION

The Contractor shall provide all labor, materials, equipment, supervision, transportation, and incidentals necessary to **furnish, install, connect, test, and commission one (1) 30 kW standby generator system** to provide emergency backup electrical power.

The generator system shall consist of a **natural gas or propane powered standby generator**, automatic transfer switch (ATS), fuel connections, electrical connections, grounding system, and startup services.

The generator shall be a **Quiet Connect or equal to rated 30 kW, 120/208V, 60 Hz**, meeting all requirements of this specification.

2. APPLICABLE CODES AND STANDARDS

All equipment and installation shall comply with the most current versions of the following:

- ****National Fire Protection Association NFPA 70 – National Electrical Code (NEC)**
- ****National Fire Protection Association NFPA 110 – Emergency and Standby Power Systems**
- ****Underwriters Laboratories UL 2200 – Generator Set Safety Standard**
- **International Building Code Seismic Requirements**
- ****U.S. Environmental Protection Agency Emergency Stationary Engine Emissions – 40 CFR 60**

All work shall meet federal, state, and local electrical and building codes.

3. GENERATOR EQUIPMENT

The contractor shall provide one **30 kW standby generator set** meeting the following minimum specifications.

Electrical

- Rated Output: **30 kW standby**
- Voltage: **208 VAC**
- Frequency: **60 Hz**
- Engine Speed: **1800 RPM**
- Voltage Regulation: **±1.0%**
- Frequency Regulation: **Isochronous**
- Random Frequency Variation: **±0.75%**

Engine

- Fuel Type: **Natural Gas or Propane Vapor (field selectable)**
- Configuration: **Inline 4-cylinder engine**
- Displacement: **2.4 liters**
- Turbocharged and aftercooled
- Electronic governor

Alternator

- 60 Hz brushless alternator
- 120°C temperature rise at 40°C ambient
- Torque-matched exciter and voltage regulator

Starting System

- 12-volt electric start
- Minimum **550 CCA battery capacity**
- 50 amp battery charging alternator
- 6 amp battery charger

4. FACTORY INSTALLED COMPONENTS

The generator shall include the following factory-installed accessories:

- Coolant heater
- Crankcase ventilation heater

- Large battery rack
 - Flexible fuel line
 - Factory muffler
 - PowerCommand® 1.1 control system or equal to
 - Full documentation package including operator, installation, and service manuals
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5. ENCLOSURE

The generator shall include a factory installed aluminum weatherproof enclosure with the following characteristics:

- Sound Level 1 acoustic enclosure
 - Corrosion-resistant aluminum construction
 - Sandstone color finish
 - Wind rating of **180 mph**
 - Integrated muffler
 - Outdoor rated
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6. AUTOMATIC TRANSFER SWITCH

The contractor shall provide and install an **automatic transfer switch (ATS)** compatible with the generator set.

Minimum ATS requirements:

- Service rated
 - 208 VAC compatible
 - UL listed
 - NEMA 3R outdoor enclosure if installed outdoors
 - Automatic start/stop generator control
 - Time delay for generator start and cool-down
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7. ELECTRICAL INSTALLATION

The contractor shall provide all wiring and electrical components necessary for complete system installation including:

- Power feeders between generator and ATS
- Control wiring
- Conduit and fittings
- Equipment grounding conductor
- Grounding electrode system

Installation shall comply with **NEC Article 700 and 702** where applicable.

8. FUEL SYSTEM CONNECTION

The contractor shall connect the generator to the existing **natural gas or propane fuel system**.

Work shall include:

- Fuel piping
 - Pressure regulator if required
 - Shutoff valves
 - Leak testing
 - Compliance with fuel supply pressure requirements:
 - Minimum: **1.5 kPa (6 in H₂O)**
 - Maximum: **3.5 kPa (14 in H₂O)**
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9. STARTUP AND COMMISSIONING

Upon installation completion, the contractor shall:

- Inspect system installation
- Perform manufacturer recommended startup
- Verify electrical connections
- Verify fuel system operation
- Perform load testing
- Verify automatic transfer operation
- Record voltage and frequency output

Startup shall be performed by a **factory-authorized technician or qualified generator technician**.

10. TESTING

Functional testing shall include:

- Generator startup and shutdown
 - Transfer switch operation
 - Automatic load transfer
 - Emergency shutdown verification
 - Minimum **30-minute operational test under load**
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11. TRAINING

The contractor shall provide on-site training for government personnel including:

- Generator operation
 - Basic maintenance procedures
 - Emergency shutdown procedures
 - Control panel operation
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12. WARRANTY

The generator shall include the manufacturer's **two (2) year limited standby warranty** covering parts and labor.